

DANIEL DESAI

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SUMMARY

Systems integration and test professional with 7+ years of experience verifying and validating complex systems for flight and autonomous operation. Qualified the PALOMINO propulsion system at Revolution Space prior to flight on Cygnus NG-23. Led end-to-end testing for Level 4 autonomous vehicle operations at Optimus Ride and supported ADAS integration during Fisker's prototype-to-production transition at Magna. Proficient in 2D and 3D CAD with GD&T, MATLAB/Simulink, Python, Julia, C++, electronic test equipment, and mechanical assembly/shop tools for rapid prototyping. Awarded U.S. Patent 11,549,787, developed for Special Operations Forces applications. U.S. Navy and Coast Guard veteran bringing experience in high-autonomy operational leadership and multidisciplinary technical implementation across aviation, land, and maritime domains.

RECENT EXPERIENCE

Senior Application Engineer

Automated Tire | Woburn, MA

July 2025 – March 2026

- Manufactured, deployed, and operated the world's only on-vehicle tire change and wheel balancing robots, resolving real-time mechanical, electrical, and software issues in a live customer environment.
- Developed a system metrics and observability application to streamline operator workflows with real-time guidance and issue logging; extended capabilities to provide battery/charging analysis and support-equipment control.

Test Engineer

Revolution Space | Boston, MA

February 2024 – January 2025

- Qualified the PALOMINO electrospray propulsion system for the Pathfinder 1 mission aboard Cygnus NG-23 by developing and executing component- and system-level vibration, shock, TVAC, and EMI/EMC testing per NASA GEVS and SpaceX Falcon payload requirements.
- Designed and prepared production drawings for mechanical ground support equipment in SolidWorks, outsourced machining, and completed final assembly.
- Authored Python (NumPy, SciPy) analysis tools to process hot fire test data and derive propulsion performance metrics.

System Test Engineer

Magna International | Boston, MA

January 2022 – May 2023

- Applied verification and validation expertise, following acquisition of Optimus Ride as the Magna Advanced Development Center, to integrate advanced driver-assistance systems into production vehicles.
- Embedded with Magna Steyr (Graz, Austria) during the Fisker program's transition from prototype to series production, supporting sample management and process optimization across international engineering teams.

System Engineer

Optimus Ride | Boston, MA

December 2018 – December 2021

- Designed and led a complete end-to-end stable release and feature testing program on city streets and a dedicated test track, validating perception, planning, and motion control performance while identifying edge cases and failure modes to determine full-system readiness for Level 4 autonomous operations.
- Acted as the on-site expert for track operations, conducting and evaluating structured tests including signalized intersection detection and response with an enhanced tester-facing HMI.
- Coordinated with software, mechatronics, and product teams to guide fixes and refinements.